ZENVIA

Hackathon Prototype Budget & Bill of Materials

Hybrid smart-home security, multimodal interaction, automation and AI analytics

PLANNING EDITION • INDIA • OCTOBER 2026

ONE-TIME PROCUREMENT PLAN	3-MONTH PLANNING ENVELOPE	BUDGET CEILING
■14,822	**■20,822**	**■25,000**
Includes 12% contingency	Includes up to ■6,000 recurring reserve	Target cap supplied for planning

Purpose

A practical cost plan for a hackathon-ready Zenvia demonstration combining physical sensors and actuators with a software dashboard, intent/automation logic, and basic analytics. The estimate assumes a laptop and smartwatch are already available and that the current stage is simulation-only; no hardware has yet been purchased.

Budget recommendation

Keep the full ■25,000 ceiling approved, but target approximately **■14,822** for one-time procurement and preparation. Hold the remaining budget for up to three months of optional cloud/API use, replacements, or late-stage integration needs rather than spending it automatically.

Important: This is a planning estimate, not a supplier quotation. Prices can change with stock, GST, shipping, seller, and exact module revision. Confirm compatibility and current prices before ordering.

1. Executive Budget Summary

Cost category	Basis	Estimate
Base BOM from supplied component guide	18 listed line items; itemized line prices summed independently	■6,774
Additional feature hardware / build allowances	Biometric, presence, environment, door sensing, feeder demo, audio, enclosure and protection	■5,660
Shipping / procurement allowance	Planning allowance; final depends on vendors and order split	■800
Pre-contingency subtotal	Base BOM + additional hardware + shipping	■13,234
12% contingency	Rounded to nearest rupee	■1,588
One-time planning total	**Target procurement / prototype preparation budget**	**■14,822**
Optional recurring reserve (3 months)	Up to ■2,000/month for optional APIs/cloud/notifications	■6,000
Three-month planning envelope	**One-time total + maximum recurring reserve**	**■20,822**
Headroom below ■25,000 cap	Uncommitted buffer after the envelope above	■4,178

What this budget covers

• Physical prototype hardware for sensing, local alerts, entry/exit sensing, a camera demo, keypad/biometric access, and a small automation/feeder demonstration.
• Software work using free/open-source tools where possible: firmware, local dashboard, MQTT messaging, local data storage and basic analytics.
• A limited optional reserve for AI API/cloud hosting and external notifications if needed for the demo.

What it does not cover

A new laptop or smartwatch, production-grade home-security installation, certified access-control equipment, medical-grade monitoring, paid developer labour, or a full-time cloud deployment. Smartwatch integration depends on the watch/app platform and available APIs; the existing watch is assumed to be owned already.

2. Base Bill of Materials

These 18 line items are transcribed from the supplied “Consolidated Bill of Materials & Component Guide” and priced using the individual line-item figures on pages 1–2. Quantities are the guide's quantities, not a recommendation to buy every item before checking what can be reused.

#	Component	Qty	Unit price	Line total
1	ESP32 DevKit V1 (30-pin)	4	■450	■1,800
2	0.96-inch I²C OLED SSD1306	3	■220	■660
3	1-channel 5V relay module	3	■80	■240
4	12V solenoid door lock	2	■400	■800
5	5V buzzer	3	■25	■75
6	Green LED	2	■5	■10
7	Red LED	3	■5	■15
8	RGB LED	1	■15	■15
9	220–330Ω resistors	7	■2	■14
10	Momentary push buttons	6	■10	■60
11	PIR motion sensor HC-SR501	1	■90	■90
12	Ultrasonic sensor HC-SR04	1	■95	■95
13	830-point breadboard	4	■140	■560
14	Jumper-wire bundles	3	■120	■360
15	Generic white sensor module (description to confirm)	1	■250	■250
16	12V 2A adapter + buck converter	1	■350	■350
17	ESP32-CAM / OV2640 camera module	2	■650	■1,300
18	4×4 matrix keypad	1	■80	■80
	Base BOM total (calculated from listed lines)			**■6,774**

Cost-source reconciliation note

The supplied guide states a core setup total of ■5,045 and a grand total of ■6,905, but its itemized prices on pages 1–2 sum to **■6,774**. Page 3 also appears to show different prices for the camera modules and the generic white sensor module. To avoid silently mixing conflicting figures, this budget uses the page 1–2 itemized prices consistently and flags the mismatch for confirmation before purchase.

Purchase optimisation

Because this is currently a simulation-only project, do not order the entire base BOM immediately. Confirm the minimum physical demo architecture first; multiple ESP32 boards, OLEDs, relay modules, locks and camera modules may be unnecessary for a single station. Reuse any available boards, wiring, power supplies, camera modules and enclosure parts.

3. Additional Hardware & Build Allowances

These are planning allowances added to support the requested broader Zenvia demonstration. They are not confirmed supplier quotes. The final selection can be reduced or substituted after the physical architecture is fixed.

Item	Qty	Unit estimate	Line estimate	Purpose / notes
AS608 fingerprint reader	1	₹1,000	₹1,000	Optional biometric authentication; verify voltage/interface before wiring.
mmWave presence sensor (e.g., LD2410C class)	1	₹410	₹410	Presence detection supplement; final model may change price.
DHT22 temperature/humidity sensor	1	₹130	₹130	Basic environment telemetry.
Magnetic door-contact sensor	2	₹60	₹120	Entry/exit state sensing; estimate per unit.
SG90 micro servo	1	₹100	₹100	Small demo latch/feeder mechanism; not a security-rated lock.
Feeding container + mounting parts	1	₹500	₹500	Allowance; adapt to chosen prototype.
Microphone/speaker accessories	1	₹700	₹700	Voice interaction demo; excludes a dedicated compute device.
Enclosure, mounting, connectors	1	₹1,200	₹1,200	Allowance for tidy, portable hackathon presentation.
Fuses, driver/protection & power accessories	1	₹1,000	₹1,000	Allowance; final needs depend on load and power design.
Spare wiring / small components	1	₹500	₹500	Allowance for replacements and iteration.
Additional hardware total			**₹5,660**	

Price sanity checks

The planning allowances were cross-checked against example online listings for comparable modules, including AS608 fingerprint readers, mmWave presence sensors, DHT22 sensors and SG90 servos / solenoid locks. Listings and stock change frequently; the exact model, seller, tax treatment and delivery cost may differ. Treat the numbers in this document as rounded budget values, not a live price guarantee.

Scope reminder

The “microphone/speaker accessories” allowance is for a voice-interaction demonstration and does not include a dedicated compute board or laptop. The “feeding container + mounting parts” line is a simple prototype allowance, not a complete certified pet-feeding product.

4. Software, AI & Recurring Costs

Service / layer	Planning cost	Approach
ESP32 firmware / Arduino IDE	■0	Free development tooling; assumes user-built firmware and no paid labour.
Local dashboard / web UI	■0	Run locally on an existing laptop or local network for the demo.
MQTT + local data store / analytics	■0	Use free/open-source broker, database and charting options where feasible.
Optional AI API / cloud hosting	■0–■1,500/month	Depends on model usage, hosting choice, traffic and free-tier availability.
Optional notifications / external integration	■0–■500/month	May be free or usage-based; provider/platform terms apply.
Conservative recurring allowance	**Up to ■2,000/month**	**Reserve, not a guaranteed expense.**

Suggested software architecture for a hackathon

Layer	Prototype responsibility
Sensors & devices	ESP32 nodes read motion, door contacts, environment and access inputs; relay/driver circuits control demo outputs.
Messaging	MQTT or a simple local API transports sensor events to the dashboard.
Rules & safety	Explicit rules check authenticated owner state, occupancy cues and door events before triggering alerts or demo actions.
AI / analytics	Summarise events, detect patterns, and provide natural-language interaction only where useful; safety-critical actions remain rule-checked.
Dashboard & alerts	Show current state, event history, alerts, basic trends and a clear demo status for judges.

Important smartwatch limitation

Owning a smartwatch does not automatically make all its readings available to Zenvia. Access depends on the watch brand, companion app, operating system permissions and vendor APIs. Present health metrics as optional integration until the exact supported data path is verified; do not describe consumer-watch readings as medical-grade diagnosis.

5. Build Plan, Safety & Procurement Controls

Phase	Deliverable	Budget gate
1 — Simulate first	Complete Wokwi firmware flow for keypad/access input, motion event, alarm state, status display and event logic.	■0 hardware spend until the simulation is reviewed.
2 — Lock the minimum demo	Choose one controller, one entry sensor, one motion/presence sensor, one alert output, and only the required camera/access modules.	Remove duplicates and confirm exact module interfaces.
3 — Buy core hardware	Purchase only confirmed compatible modules and essential wiring/power protection.	Target the itemised BOM; compare vendor totals including delivery.
4 — Add AI & dashboard	Build local dashboard, event history, rule engine and analytics; add cloud/API only if it materially improves the demo.	Use free/local stack first; track any monthly usage.
5 — Demo hardening	Test false alarms, sensor failures, power interruptions, owner/visitor states and a manual safe override.	Use contingency only for documented gaps or failures.

Electrical and physical safety

• Never power a 12V solenoid lock or other high-current load directly from an ESP32 GPIO pin. Use a suitable driver/MOSFET or rated relay module and an appropriate supply.
• Use a flyback diode or suitable suppression across inductive DC loads where required by the driver design; verify the relay/driver module's protection before wiring.
• Use the correct voltage for every module, common ground where the circuit design requires it, and suitable fuses/current limits. Do not expose mains wiring on a student prototype.
• A servo-based feeder/latch is for demonstration only. Do not rely on the prototype as the sole means of securing a real home or protecting people/pets.

Procurement checklist

■ Confirm exact ESP32 board, camera, sensor and keypad variants.

■ Check whether any items are already available or can be borrowed.

■ Verify vendor price, tax, shipping, delivery time and return policy.

■ Check power requirements and pin compatibility before ordering.

■ Keep invoices and update actual spend against the ■25,000 ceiling.

6. Assumptions & Notes

• Location / currency: India; all values in Indian rupees (INR).
• Current state: Wokwi simulation only; no hardware purchased yet.
• Existing assets: laptop and smartwatch assumed available at no incremental cost.
• Price status: base BOM values originate from the user-supplied component guide; additional items, shipping, cloud/API use and contingency are planning estimates.
• Contingency: 12% of the pre-contingency subtotal, rounded to ■1,588.
• Three-month envelope: uses the upper recurring allowance of ■2,000/month, not a prediction that all recurring charges will occur.
• Budget headroom: ■4,178 remains under the ■25,000 cap after the full three-month envelope.

Decision summary: Use the ■25,000 cap as the maximum authorised envelope, but stage purchases. First validate the design in Wokwi, then purchase the minimum compatible hardware set, prioritising a reliable security demo and a clear dashboard over adding every possible feature at once.

Prepared as a hackathon planning document. Estimates should be refreshed before procurement; this document is not a supplier quote or a production-security specification.